Stat 5200, Module 3: Completely Randomized Designs and ANOVA

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Overview

Objectives and overview

- This material is based Chapter 3 on the textbook “A First Course in Design and Analysis of Experiments” (Oehlert, 2000).
- Chapter outcomes include:
  - TODO

Completely Randomized Designs

Completely Randomized Design

- Completely randomized design (CRD) is the simplest design for comparing treatments and is sometimes the preferred design.
- Setup:
    - There are g treatment groups.
    - N experimental units (EU)s available.
    - You want sample sizes of $n_1, n_2, \ldots, n_g$, with $\sum_{i=1}^{g} n_i = N$.

Completely Randomized Design II

- Idea: All possible sub-group composition are equally as likely.

Performing a Completely Randomized Design

- Randomly choose n_1 EU's for treatment 1.
- Randomly choose n_2 EU's for treatment 2.
- $\dots$
- Randomly choose n_{g-1} EU's for treatment $g-1$.
- The remaining n_g EU's are assigned to treatment g.

- On a computer, this can be done by permuting $1, 2, \dots, N$, and assigning the first n_1 numbers to treatment 1, the next n_2 to treatment 2, etc.

Advantages and Disadvantages

- Advantages
  - Simple to do; all you need is one random permutation.
  - Simple to explain and understand.
  - Simple to analyze.
  - No restrictions on the sample sizes (other designs may restrict n_i).
  - Easy with missing values.
- Disadvantages
  - Not always possible or cost-effective.
  - The accuracy of the experiment depends on variability among the EUs, not just the variability among treatment groups. Thus, it may require more EUs ($$$$).

Example: Acid Rain

Wood and Bormann (1974) studied the effect of acid rain on trees. One experiment used 240 six-week-old yellow birch seedlings. These seedlings were divided into five groups of 48 *at random*, and the seedlings within each group received an acid mist treatment 6 hours a week for 17 weeks. The five treatments differed by mist pH: 4.7, 4.0, 3.3, 3.0, and 2.3. Othwerwise, the seedlings were treated identically. After the 17 weeks, the seedlings were weighed, and total plant (dry) weight was taken as response.

Questions:

- What type of design is this, and can you describe why?
- What are the EUs, and how many are there?
- What was the response?

Example: Acid Rain II

- What compromises may have been made in this case?

Like many experiments, the researchers necessarily have to narrow down an enormous question to a workable experiment using artificial acid rain under controlled conditions. For any experiment, a considerable amount of non-statistical background work goes into the planning even the simplest experiment.

Models and Parameters

- All formal statistical analysis is based on probability models.
- *EX:* The number of heads in ten tosses of a fair coin would be Binomial$(10, 0.5)$. Here, the model depends on 2 parameters: $N = 10$ the number of trials, $p = 0.5$ is the probability of heads.
- In an experiment, we may posit several different models for the data, all with *unknown* parameters. One of our objectives is often to describe which model is the best description of the data, and make *inference* about parameters.

Definition: Inference

Using data to learn properties of a probability distribution. In our case, using data to estimate parameters of a model.

Introduction and definitions II

- The models we will consider for experimental design will have two basic parts:
  - A description of the expected value, or mean, of the data. Sometimes called “model for the means”.
  - A description of how data vary around the treatment means. Sometimes called ‘model for the errors”.
- For the errors, we will generally assume that they are independent for different data values, have mean zero, and all have a common variance σ^2.
- To perform formal tests and obtain confidence intervals, we will further assume a distribution of the errors. Generally, we assume that the errors follow a Gaussian (Normal) distribution.

Introduction and definitions III

- For a CRD, we start with two basic models.
- Recall: g treatment groups, N EUs.

1. The Full Model (separate group means)

Introduction and definitions IV

2. The Reduced Model (single mean model)

Defining overall mean

- It's sometimes useful to express the group means μ_i as:

$$\mu_i = \mu^* + \alpha_i.$$

  In this case, we call μ^* the *overall mean*, and the α_i's are the ith *treatment effects*.
- This introduces an extra parameter (g+1 total), which we need to address. Generally, we do this by defining μ^* and require that $\alpha_i = \mu_i - \mu^*$.

Defining overall mean II

- One classic choice is to define μ^* as the mean of the treatment means:

Defining overall mean III

- An alternative choice (that can sometimes make hand-work simpler) instead describes μ^* as the weighted average:

Defining overall mean IV

- Another common choice in software is to set $\mu^* = \mu_j$, for some $j \in 1 : g$. Many software just use the last mean: $\mu^* = \mu_g$.

Model Recap

However you choose to define μ^* and α_i's, we end up with the following models for the data:

- *The full model*:

$$y_{ij} = \mu^* + \alpha_i + \epsilon_{ij}$$

- *The reduced model*:

$$y_{ij} = \mu + \epsilon_{ij}$$

Parameter Estimation

Estimating parameters

- We want to use data to estimate μ_i's and the variance σ^2. From these, we'll compute μ^* and α_i's.
- We will focus on computing unbiased estimates of these parameters.
- Estimate for μ_i:
- The estimate for μ^* is fixed, so it depends on how we defined it.

Estimating parameters II

- Unweighted average

- Weighted average

- Reference Group

Estimating parameters III

- The only parameter left to estimate is σ^2:

Estimating parameters IV

- Above, we have described point estimates. In some sense, these are our "best guess" as to the value of a particular parameter.
- We also want some measure of uncertainty of this estimate. The standard frequentist way of doing this is a confidence interval.
- Confidence intervals for α are not very useful, because it depends on how we define them. Confidence intervals for group means are typically defined as:

Comparing Models: ANOVA

Analysis of Variance

- In our analysis of a CRD, our models focus on the *mean responses* of the treatment groups.
- A basic question we might want answered: Are all the means the same, or are some different?
- We can express this same question mathematically:

Analysis of Variance II

- The basic idea for many statistical tests (including ANOVA) follows *Occam's Razor*: we use the simplest model that is consistent with the data; we will only use a more complicated model if the data indicate that the more complicated model is needed.
- How do we know if a more complex model is needed? We compare residuals.

Analysis of Variance III

- The lower the SSR, the more of the variance in the data is explained by the model.
- Importantly, the SSR for the reduced (single mean) model can *never* be lower than the separate model, and in practice, is nearly always larger.
- What we need now is a way to consistently decide when the separate means model is sufficiently "better" than the single means model.
- To do so, we will set up a statistical hypothesis test:

ANOVA mechanics

- ANOVA works by partitioning the total variance in the data into parts that mimic the model.
- For instance, in the separate mean model, the model says the data differ from the grand mean by a random error and a treatment effect:

  ANOVA will have a similar decomposition: one part dealing with group means, one part with deviations from those means:

ANOVA mechanics II

- Recall from Eq. **??**, we have:

$$y_{ij} - \bar{y}_{\bullet\bullet} = \hat{\alpha}_i + r_{ij},$$

ANOVA mechanics III

- Finally, we can connect this back to the residuals.
- The Sum of Squared Residuals (SSR) depended on our model:

$$SSR_{\text{single means}} = \sum_{i=1}^{g}\sum_{j=1}^{n_i}\left(y_{ij} - \bar{y}_{\bullet\bullet}\right)^2 = SS_T.$$

$$SSR_{\text{separate means}} = \sum_{i=1}^{g}\sum_{j=1}^{n_i}\left(y_{ij} - \bar{y}_{i\bullet}\right) = SS_E$$

Calibrating Effect Size

- We have established:
  - We will compare single mean against separate mean by comparing their SSRs.
  - Comparing SSR can be done by decomposing total error into two parts: $SS_T = SS_{Trt} + SS_E$, and if the separate model is "much better" than the SS_{Trt} will be large.
- **How large is large enough?**

Calibrating Effect Size II

- The answer also lies in the decomposition of sum of squares.

ANOVA variable recap

- Sum of Square Residuals:

$$SSR = \sum_{i=1}^{g} \sum_{j=1}^{n_i} \left(y_{ij} - \hat{y}_{ij} \right)^2$$

- Treatment Sum of Squares:

$$SS_{Trt} = \sum_{i=1}^{g} \sum_{j=1}^{n_i} \left(\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet} \right)^2 = \sum_{i=1}^{g} n_i \hat{\alpha}_i^2 .$$

- Error Sum of Squares:

$$SS_E = \sum_{i=1}^{g} \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i\bullet})^2$$

ANOVA variable recap II

- Total Sum of Squares:

$$SS_T = SS_{Trt} + S_E = \sum_{i=1}^{g} \sum_{j=1}^{n_i} \left(y_{ij} - \bar{y}_{\bullet\bullet}\right)^2$$

- F-statistic:

$$\frac{SS_{Trt}/(g-1)}{SS_E/(N-g)} \sim F_{g-1,N-g}.$$

Contrasts

Introduction to Contrasts

- An ANOVA can give us a good overview as to whether all the treatment groups have the same mean response.
- However, an ANOVA by itself does not tell which treatments are different or in what ways they differ.
  - One of the problems: There are many ways to define treatment effects, so confidence intervals are not immediately useful.
- One method to examine treatment effects is called a contrast.

Formal Definition

Definition: Contrasts

A contrast is a linear combination of treatment means, μ_i's:

$$W(\mu_{1:g}) = \sum_i w_i \mu_i, \quad \text{with} \quad \sum_i w_i = 0$$

- The condition $\sum_i w_i = 0$ guarantees that $W(\mu_{1:g})$ does not depend on μ.

Formal Definition II

- Often the weights w_i themselves are referred to as a contrast.
- An observed contrast or estimated contrast is defined as:

$$W\left(\bar{y}_{1:g,\bullet}\right) = \sum_i w_i \bar{y}_{i\bullet} = \sum_i w_i \hat{\alpha}_i = W\left(\hat{\alpha}_{1:g}\right)$$

Types of Common Contrasts

Pairwise Contrasts

- It is often of interest to compare all pairs of groups.

Types of Common Contrasts II

Comparison with a Control

- Let's suppose treatment 1 is a control.
- An obvious idea is to compare the mean or effect of the control with the average mean or effect of all non-controls.

Types of Common Contrasts III

Multiple Treatments (Factorial Designs)

- Suppose there are two treatment factors, A and B, each at two levels, 1 and 2.
- We'll denote the treatment group means as $\mu_{A1}, \mu_{A2}, \mu_{B1}, \mu_{B2}$, respectively.
- We might want to compare the averages of just one treatment type (A vs B or 1 vs 2).

Types of Common Contrasts IV

- What do we do with contrasts?
- If a contrast is large (in magnitude), that provides evidence that there is a difference between treatment levels that are specified by the contrast. Conversely, if it is small, there is little evidence of a difference.
- The common question arises: how large is "large enough" to declare the contrast is not zero?
- We can use the linearity of the model to answer this question!

Inference on Contrasts

- Under the null hypothesis and normality assumption, we can use Student's t with $df = N - g$ in tests or confidence intervals.
- Under the null hypothesis, $W(\mu_i) = \sum_i w_i \mu_i = 0$.
- The standardized estimated contrast follows a student-t distribution with $df = N - g$:

$$t = \frac{\sum_i w_i \bar{y}_{i\bullet}}{\sqrt{MS_E}\sqrt{\sum_i w_i^2 / n_i}}, \quad MS_E = \frac{\sum_{i=1}^{g} \sum_{j=1}^{n_i} \left(y_{ij} - \bar{y}_{\bullet\bullet}\right)^2}{N - g}$$

Inference on Contrasts II

- A $1-\alpha$ confidence interval for a contrast $W(\mu_i)$ has the usual form:

$$\text{estimate} \pm \text{critical value} \times \text{standard error}$$

Multiple Comparisons

The Need for Multiple Comparisons

- When picking contrasts, it is important to consider the issue of multiple comparisons.
- Idea: we pick a α value to test at (e.g., $\alpha = 0.05$). That means, for a single test, there is only a 5% chance $(1/20)$ that we will "incorrectly reject" a true null. **What if we conduct 15 tests?**

The Need for Multiple Comparisons II

- A test of a single hypothesis is characterized by two error rates: the Type I error rate (α) and the Type II error rate (β).

Types of Error Rates

When there are K hypotheses, the situation becomes more complicated, and we must define which type of error rate we are trying to control.

- Per Comparison Error Rate: The probability of a Type I error relative to any single, specific hypothesis. In the context of multiple comparisons, this is just our standard α.
- Familywise Error Rate (FWER): The probability of rejecting at least one true null hypothesis when all null hypotheses are true. This is also referred to as the experimentwise error rate.
- Strong Familywise Error Rate: The probability of making any false discoveries (rejecting at least one true H_0), regardless of how many correct rejections are made.

Types of Error Rates II

- False Discovery Rate (FDR): The expected proportion of false discoveries among all the rejected hypotheses.

Bonferroni Method

- The Bonferroni method relies on the Bonferroni Inequality, which states that for K events, $P(E_1 \cup E_2 \cup \ldots \cup E_K) \leq \sum_{i=1}^{K} P(E_i)$.
- If you test K true null hypotheses at level α, the probability of rejecting one or more is bounded by $K\alpha$.
- To control the familywise error rate at α for K simultaneous contrasts, the Bonferroni method uses α/K as the significance level for each individual comparison.
- Equivalently, you can multiply each ordinary P-value by K to obtain a "Bonferronized P-value".

Bonferroni Method II

- **Usage Rule:** The Bonferroni method is designed for a family of K contrasts chosen in advance of seeing the data. When you are testing 6 or fewer pre-chosen contrasts, Bonferroni is better than HSD (Tukey).

Tukey's Honestly Significant Difference (HSD)

- Tukey's HSD is based on the Studentized range distribution (Q), which is characterized by the number of groups g and the degrees of freedom for error, independent of μ or σ.
- For equal sample sizes, we reject the null hypothesis if the maximum mean minus the minimum mean is greater than or equal to the HSD.
- The formula is: $HSD = q_{1-\alpha}(g, N-g)\frac{s_p}{\sqrt{n}}$.
- **Unequal Sample Sizes:** Its disadvantage is that it assumes equal sample sizes. When this is violated, the extension of Tukey-Cramer method can be considered.

Tukey's Honestly Significant Difference (HSD) II

- **Usage Rule:** HSD (Tukey) is designed just for pairwise comparisons, although it can be extended to all contrasts. Most often, HSD is the preferred method.

Scheffé Method and Simultaneous Confidence Intervals

- **Scheffé Method**: Controls the familywise error rate for the infinite set of all contrasts.
- The constant used in the method is computed based on F distribution, and is called Scheffe Significant Difference (SSD).
- **Usage Rule:** There are only two situations in which you should consider the Scheffe method:
  1. You want to test K preplanned contrasts where K is substantially greater than the number of pairwise contrasts.
  2. You want to test one or more contrasts selected after seeing the data.

Scheffé Method and Simultaneous Confidence Intervals II

- **Simultaneous Confidence Intervals**: A set of intervals $I_1, I_2, \ldots, I_k$ are $1 - \epsilon$ level simultaneous confidence intervals if the probability that *all* intervals cover the true parameters is $\geq 1 - \epsilon$.

Summary: Choosing a Method

- None of these methods is "best" and they should look to see what level of agreement there is among the methods.
- If they disagree, then it is up to the experimenter to decide the level of comfort with controlling Type I error or getting more power.
- **General Guidelines:**
    - **Few Pre-planned Contrasts (≤ 6):** Bonferroni is better than HSD (Tukey).
    - **All Pairwise Comparisons:** Most often, HSD is the preferred method (use Tukey-Cramer if sample sizes are unequal).
    - **Data Snooping / Many Contrasts:** Scheffé method aims at controlling the familywise error rate for the infinite set of all contrasts.

Summary: Choosing a Method II

- *Note: If you only have one or two pre-planned orthogonal contrasts, some practitioners choose not to adjust for multiple comparisons at all, accepting the per comparison error rate. However, controlling the familywise rate remains the safer statistical practice.*

Applied Example: Plant Growth

Example 1: Data Setup and the F-Test

- We will use R's built-in `PlantGrowth` dataset, measuring dried plant yields under a control and two different treatment conditions.
- As discussed, a Completely Randomized Design (CRD) is the simplest design for comparing treatments. We have $g = 3$ groups and $N = 30$ total plants.

Example 1: Data Setup and the F-Test II

```r
# Load data and check the structure
data(PlantGrowth)
str(PlantGrowth)
```

```
'data.frame': 30 obs. of  2 variables:
 $ weight: num  4.17 5.58 5.18 6.11 4.5 4.61 5.17 4.5..
 $ group : Factor w/ 3 levels "ctrl","trt1",..: 1 1 1..
```

```r
# Ensure the treatment is a factor (categorical)
class(PlantGrowth$group)
```

```
[1] "factor"
```

Example 1: Data Setup and the F-Test III

- We first test the null hypothesis that all group means are equal using an omnibus ANOVA F-test.

```
# Run the ANOVA model
pg_model <- aov(weight ~ group, data = PlantGrowth)

# View the ANOVA table
summary(pg_model)


            Df Sum Sq Mean Sq F value Pr(>F)
group        2  3.766  1.8832   4.846 0.0159 *
Residuals   27 10.492  0.3886
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Example 1: Data Setup and the F-Test IV

- **Interpretation:** With a p-value of 0.0159, we reject the null hypothesis. There is a significant difference in plant yields between at least some of the groups. But which ones?

Example 2: Formulating Custom Contrasts

- We know from our model that the single mean model is inadequate, so we want to estimate the separate group means.
- Let's test a specific contrast: Is the control group significantly different from the average of the two experimental treatments?

Example 2: Formulating Custom Contrasts II

```r
library(emmeans)

# Get estimated marginal means
pg_emm <- emmeans(pg_model, ~ group)

# Define contrast: Control (1) vs Average of Trt1 (2) and Trt2 (3)
# Weights must sum to zero
custom_list <- list("Control vs Average Trt" = c(1, -0.5, -0.5))

# Test the contrast
contrast(pg_emm, method = custom_list)
```

```
 contrast               estimate    SE df t.ratio
 Control vs Average Trt  -0.0615 0.241 27  -0.255
 p.value
  0.8009
```

Example 2: Formulating Custom Contrasts III

- **Interpretation:** The p-value is 0.2644. We do not have sufficient evidence to say the control is different from the *average* of the two treatments.

Example 3: Multiple Comparisons

- What if we want to compare all individual pairs of treatments?
- We must adjust for multiple comparisons to control the familywise error rate. Tukey's HSD is the preferred method for all pairwise comparisons.

```
# Run all pairwise comparisons with Tukey adjustment
pg_pairwise <- contrast(pg_emm, method = "pairwise")

summary(pg_pairwise)


 contrast    estimate    SE df t.ratio p.value
 ctrl - trt1    0.371 0.279 27   1.331  0.3909
 ctrl - trt2   -0.494 0.279 27  -1.772  0.1980
 trt1 - trt2   -0.865 0.279 27  -3.103  0.0120

P value adjustment: tukey method for comparing a family of 3 estimates
```

Example 3: Multiple Comparisons II

- **Interpretation:** After applying the Tukey penalty, only the difference between `trt1` and `trt2` is statistically significant ($p = 0.0120$). The control is not significantly different from either individual treatment.

Other Common Scenarios (See Textbook)

- While Bonferroni, Tukey, and Scheffé cover the majority of basic experimental designs, there are specialized methods for other common scenarios.
- **Comparing to a Control:** If your experiment has one baseline control group and several treatments, you rarely need all pairwise comparisons. **Dunnett's Test** is specifically designed to control the familywise error rate for this exact scenario, providing more power than Bonferroni or Tukey.

Other Common Scenarios (See Textbook) II

- **High-Throughput Testing (FDR):** If you are running hundreds or thousands of tests (e.g., genomics), controlling the FWER is too strict. Methods like the **Benjamini-Hochberg procedure** control the False Discovery Rate instead.
- **Sequential / Step-Down Methods:** Procedures like **Holm's method** provide a uniformly more powerful alternative to the standard Bonferroni correction by evaluating p-values sequentially.
- **Further Reading:** We will not cover the mathematical derivations of these methods in class, but you can find their details, applications, and statistical tables in Oehlert (2000).

Testing Model Assumptions

Core Assumptions of the Errors

- The validity of ANOVA results, including contrasts and some forms of multiple testing, depends on our model assumptions.
- For the one-way ANOVA CRD model: $y_{ij} = \mu^* + \alpha_i + \epsilon_{ij}$.
- Our main assumptions for this model include:
  - All ϵ_{ij} are independent
  - The variances of ϵ_{ij}'s are all the same.
  - The ϵ_{ij}'s are normally distributed.

Core Assumptions of the Errors II

- Listing the assumptions in decreasing order of importance:
  1. **Independence** (most important)
     .
  2. **Constant variance** (next most important).
  3. **Normal distribution** (least important).

Diagnosing Violations

- Essentially all methods for diagnosing violation of assumptions are based on study of the ________________.
- Conceptually: $r_{ij} =$ ______________ $-$ ______________.
- **Tests vs. Plots:** There are tests for normality and equal variances, but they have limitations (small sample sizes lack power; large sample sizes detect trivial problems).
- Therefore, formal tests are ________________, while plots are ________________.

R Code: Principal Plots for Diagnostics

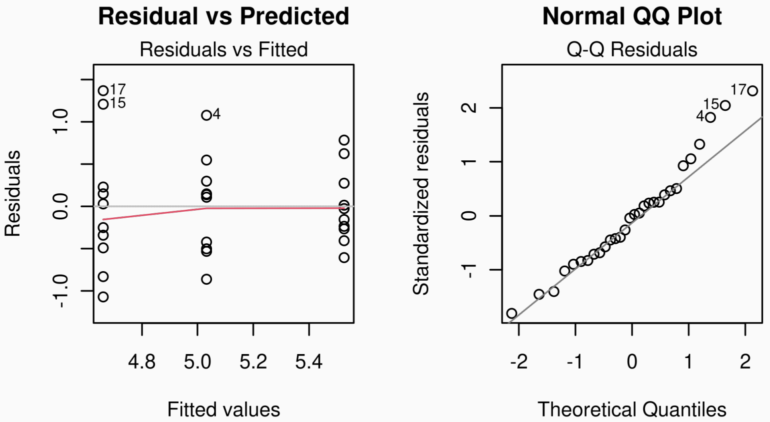
- We can generate the principal diagnostic plots (Residual vs Predicted and Normal QQ Plot) directly from our ANOVA model object in R.

```r
# Using the pg_model (PlantGrowth) from earlier
par(mfrow = c(1, 2)) # Put plots side-by-side

# 1. Residual vs Predicted (Fitted) plot for constant variance
plot(pg_model, which = 1, main = "Residual vs Predicted")

# 2. Normal probability (QQ) plot for normality
plot(pg_model, which = 2, main = "Normal QQ Plot")
```

R Code: Principal Plots for Diagnostics II



R Code: Principal Plots for Diagnostics III

```r
par(mfrow = c(1, 1))
```

- **Supplemental Tests:** If you need formal statistical tests, you can use the Shapiro-Wilk test for normality and Levene's test for equal variances.

R Code: Principal Plots for Diagnostics IV

```
# Supplemental Test for Normality on residuals
shapiro.test(resid(pg_model))



 Shapiro-Wilk normality test

data:  resid(pg_model)
W = 0.96607, p-value = 0.4379


# Supplemental Test for Equal Variances (requires 'car' package)
library(car)
leveneTest(weight ~ group, data = PlantGrowth)


Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  2  1.1192 0.3412
      27
```

Checking Specific Assumptions

- **Variance and Normality:** Heteroskedasticity (changing variances) and non-normality usually come hand-in-hand. Heteroskedasticity is more important and takes priority for remedies.
- **Independence:** Independence of the ϵ_{ij}'s cannot be checked by ____________________.
- Proper randomization is the best protection against lack of independence.

Remedies: Data Transformations

- The principle remedial tool is re-expression (transformation) of the response.
- If two transformations $\tilde{y}_1 = f_1(y)$ and $\tilde{y}_2 = f_2(y)$ have a linear relationship ($\tilde{y}_2 = \frac{\tilde{y}_1 - a}{b}$), they are completely equivalent for correcting assumptions.
- For log transformations, the base ______________________.
- The Box-Cox power family of transformations for a positive response variable is defined as:

  ____________________________.

References and Acknowledgements

Oehlert GW (2000). *A First Course in Design and Analysis of Experiments*. W. H. Freeman. ISBN 0-7167-3510-5.

Wood T, Bormann F (1974). "The effects of an artificial acid mist upon the growth of Betula alleghaniensis Britt." *Environmental Pollution (1970)*, **7**(4), 259–268.

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References and Acknowledgements II

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